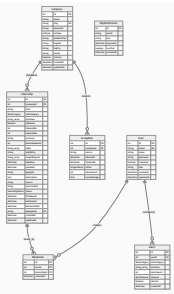
04. Database Schema & Data Models

1. Relational Database Architecture

The **Internship Tracker** data layer is powered by **PostgreSQL** (hosted serverless on Neon with AWS connection pooling and SSL encryption) managed through **Prisma ORM**.

The schema is normalized to 3NF (Third Normal Form) across enterprise profiles, role requisitions, user authentication, and alert subscriptions.

2. Entity-Relationship Diagram (ERD)



---

3. Data Dictionary & Model Specifications

A. Model `Company`

Stores company identities, corporate headquarters, business sector, and ATS tracking metadata.

Column	Type	Constraints	Description
`id`	`Int`	Primary Key, Autoincrement	Unique company identifier.
`name`	`String`	Required	Official company name (e.g. "Google", "Amazon").
`slug`	`String`	Unique, Required	URL-safe slug (e.g. "google", "tower-research-capital").
`careerUrl`	`String`	Required	Company official top-level careers landing page.
`atsType`	`AtsType`	Enum, Required	ATS platform type (`GREENHOUSE` , `LEVER` , `ASHBY` , `CUSTOM`).
`atsIdentifier`	`String`	Required	Slug identifier used in the company's ATS board URL.
`logoUrl`	`String?`	Optional	CDN URL to high-resolution vector or PNG corporate logo.
`hqCity`	`String?`	Optional	Indian corporate hub (e.g. "Bengaluru", "Gurugram").
`sector`	`String?`	Optional	Industry classification (e.g. "Big Tech", "Quant / FinTech").
`isActive`	`Boolean`	Default: `true`	Indicates whether the automated scraper monitors this company.

B. Model `Internship`

The primary table storing verified internship requisitions.

Column	Type	Constraints	Description
`id`	`Int`	Primary Key, Autoincrement	Unique internship ID.
`companyId`	`Int`	Foreign Key -> `Company.id`	Reference to parent company.
`title`	`String`	Required	Official job title (e.g. "Software Development Engineer Intern").
`roleCategory`	`RoleCategory`	Enum, Indexed	Technical category: `SDE`, `DATA`, `AI_ML`, or `OTHER`.
`locations`	`String[]`	Array	Locations where the role is offered (e.g. `["Bengaluru", "India"]`).
`isRemote`	`Boolean`	Default: `false`	True if explicitly designated as a remote/WFA position.
`stipendMin`	`Int?`	Optional	Minimum monthly stipend in INR (e.g. `110000`).
`stipendMax`	`Int?`	Optional	Maximum monthly stipend in INR (e.g. `130000`).
`currency`	`String`	Default: `"INR"`	ISO currency code.
`durationMonths`	`Int?`	Optional	Internship duration (enforced strictly to 2 or 3 months).
`skills`	`String[]`	Array	Required skills/technologies (e.g. `["PyTorch", "C++", "CUDA"]`).
`eligibility`	`String?`	Optional	Text explanation of eligible graduation batches and degrees.
`targetDegrees`	`String[]`	Array	Eligible degrees (e.g. `["BTECH", "MTECH", "MCA"]`).
`deadline`	`DateTime?`	Optional	Application deadline timestamp.
`postedAt`	`DateTime?`	Optional	Original date/time requisition was published by employer.
`applyUrl`	`String`	Required	**100% verified direct link** to the application form.
`description`	`Text`	Required	Full text job description, requirements, and role expectations.

Column	Type	Constraints	Description
`source`	`String`	Required	Origin ATS or crawler tag (`GREENHOUSE` , `LEVER` , `ASHBY` , etc.).
`sourceJobId`	`String`	Required	Unique requisition ID from employer's ATS (e.g. `10541345`).
`status`	`InternshipStatus`	Default: `OPEN` , Indexed	Listing state: `OPEN` or `CLOSED` .
`firstSeenAt`	`DateTime`	Default: `now()`	Timestamp when the role was first discovered by scraper.
`lastSeenAt`	`DateTime`	Default: `now()`	Timestamp when the scraper last confirmed existence at source.
`lastVerifiedAt`	`DateTime`	Default: `now()`	Timestamp of the most recent successful HTTP GET 200 health check.
`dedupHash`	`String`	Unique, Indexed	SHA-256 hash preventing duplicate listings.

C. Model `User`

Manages authenticated candidate accounts, saved bookmarks, and custom alerts.

Column	Type	Constraints	Description
`id`	`Int`	Primary Key, Autoincrement	Unique user account ID.
`email`	`String`	Unique, Required	Verified student email address.
`name`	`String?`	Optional	Candidate full name.
`password`	`String?`	Optional	Salted bcrypt hash (null if user authenticated solely via OTP).
`isVerified`	`Boolean`	Default: `false`	Email verification flag.
`role`	`String`	Default: `"USER"`	Access control level (`USER` or `ADMIN`).

D. Model `OtpVerification`

Stores cryptographically random one-time passwords for passwordless email verification.

Column	Type	Constraints	Description
`id`	`Int`	Primary Key, Autoincrement	Unique OTP record ID.
`email`	`String`	Indexed	Target recipient email address.
`otp`	`String`	Required	6-digit numeric OTP string.
`expiresAt`	`DateTime`	Required	Expiration timestamp (strictly 10 minutes from creation).
`verified`	`Boolean`	Default: `false`	True once successfully exchanged for a JWT session.

E. Model `Bookmark`

Stores saved internships for user candidate dashboards. Enforces a composite unique constraint `@@unique([userId, internshipId])` to prevent duplicate bookmarks.

---

4. Database Indexing & Performance Tuning

High-concurrency query performance is guaranteed through composite indexes defined in `schema.prisma`:

```
model Internship {
  ...
  @@index([companyId])
  @@index([roleCategory])
  @@index([status])
  @@index([dedupHash])
}
```

Why These Indexes Matter (Interview Talking Points):

1. **Filtering by Status (`status`)**:
   95% of queries filter for `WHERE status = 'OPEN'`. An index on `status` enables the PostgreSQL query planner to perform an **Index Scan** instead of an expensive **Sequential Table Scan** across thousands of closed/historical records.
2. **Composite Filtering (`companyId`, `roleCategory`)**:
   Enables sub-10ms lookups when students filter by category (e.g. `WHERE roleCategory = 'AI_ML' AND status = 'OPEN'`).
3. **Unique Hash Index (`dedupHash`)**:
   Underneath the hood, PostgreSQL maintains a B-Tree index on `dedupHash`. Ingestion upserts execute in `O(log N)` time, instantly detecting whether a scraped job is a duplicate without reading the entire table.